DATA ANALYSIS · APRIL 2026

Domestic Violence & Socioeconomic Factors

A statistical exploration of how education, employment, income, and marital status relate to domestic violence risk.

347	5	1
OBSERVATIONS	PREDICTORS	BINARY OUTCOME

github.com/cocobutterx13/Domestic-Violence-Analysis

OVERVIEW

Why It Matters

Domestic violence disproportionately affects women across all demographics. This project investigates whether socioeconomic factors (education, employment, income, and marital status) can predict the likelihood of experiencing domestic violence.

KEY RESEARCH QUESTIONS

01 How do socioeconomic factors (education, employment, income) relate to domestic violence vulnerability?

02 Does marital status influence exposure to or experience of domestic violence?

03 Which demographic and contextual factors are most predictive of vulnerability when modelled jointly?

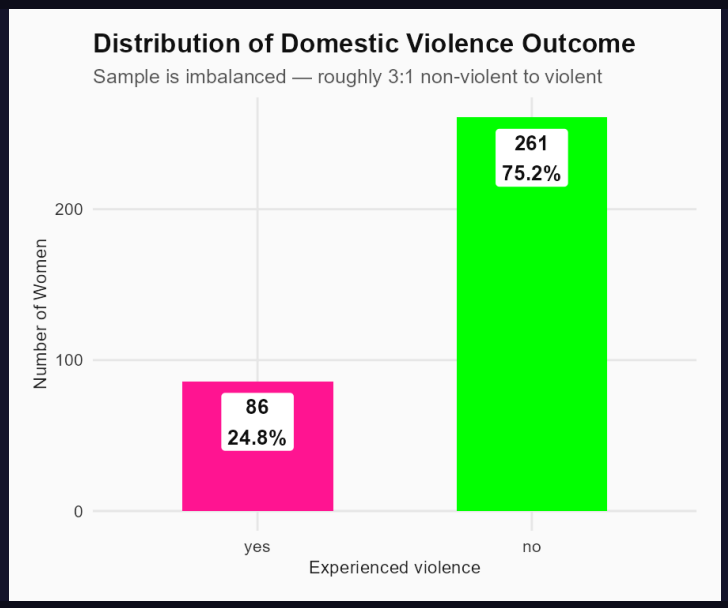
DATASET

347 observations · Variables: Age, Education, Employment, Income, Marital Status → Violence (yes/no)

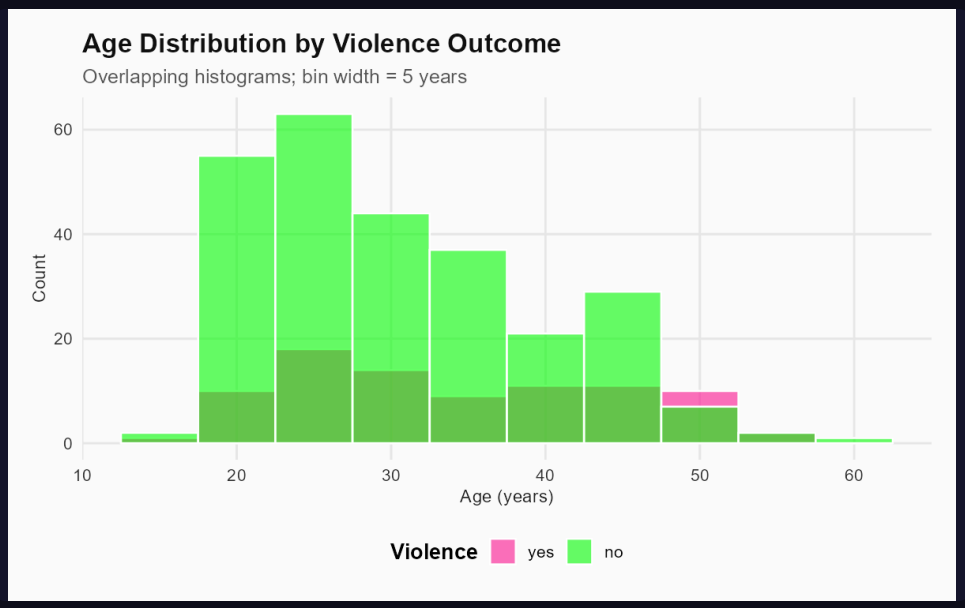
EXPLORATORY ANALYSIS

Class Balance & Age

The dataset is imbalanced. This matters for model sensitivity and accuracy interpretation.



Outcome Distribution



Age by Violence Group

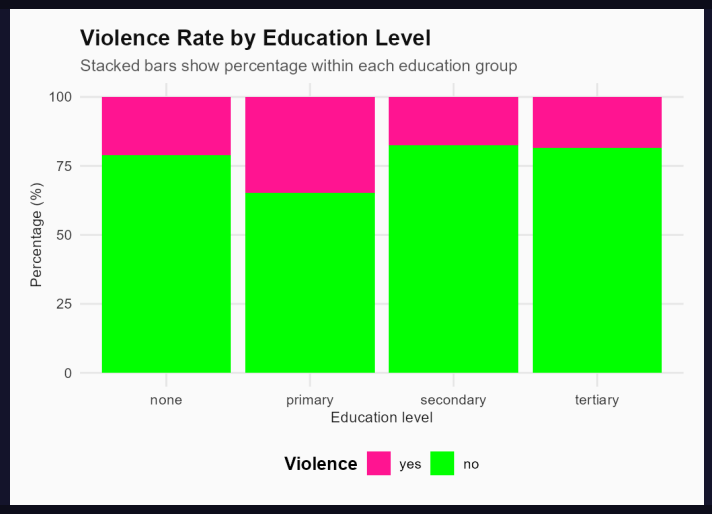
About 75% of the sample reported no violence (≈3:1 imbalance). This inflates overall accuracy — sensitivity is the more meaningful metric.

Age distributions overlap substantially. The difference is statistically significant, but small in magnitude, suggesting age is not a dominant predictor alone.

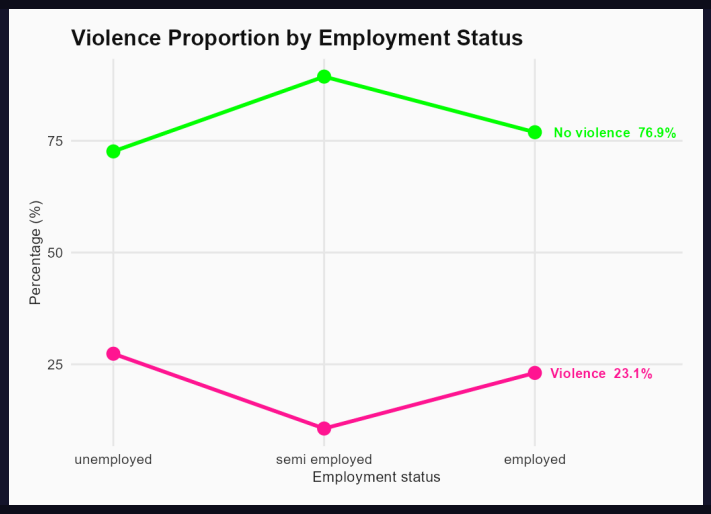
CATEGORICAL PREDICTORS

Education & Employment

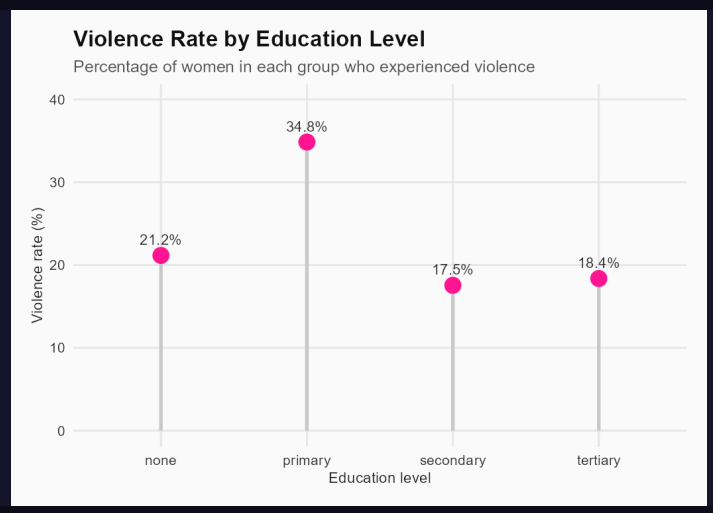
X^2 P < 0.05 | SIGNIFICANT PREDICTORS



Education vs Violence



Employment vs Violence



Education Lollipop

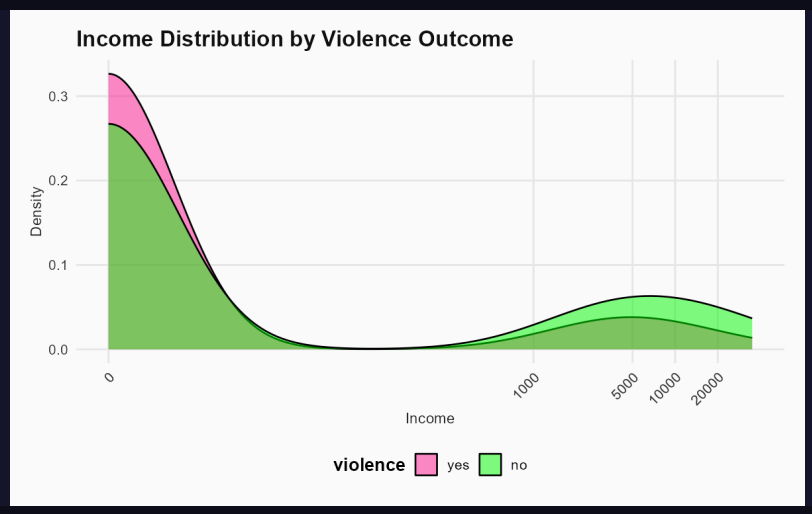
Violence rates vary non-linearly across education levels, with primary education showing the highest risk.

The lollipop chart highlights relative proportions clearly; primary education stands out as the highest-risk group.

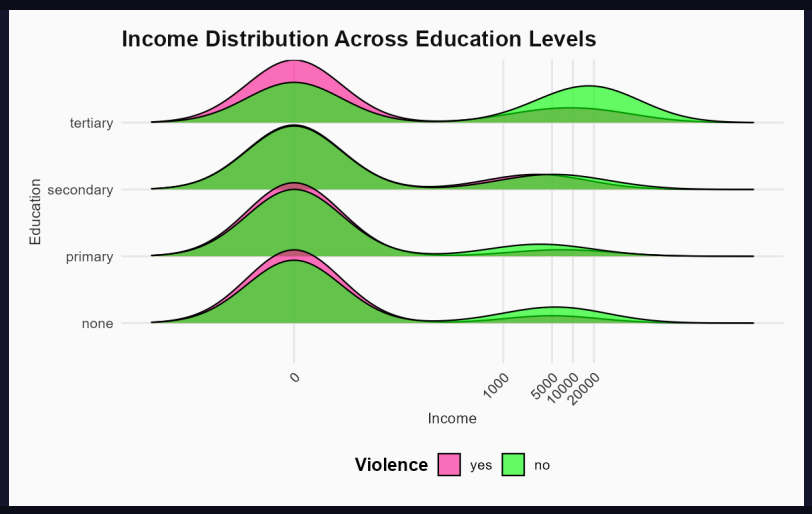
Unemployed respondents face the highest violence rates. Employment provides economic independence and reduces vulnerability.

Income: A Complex Picture

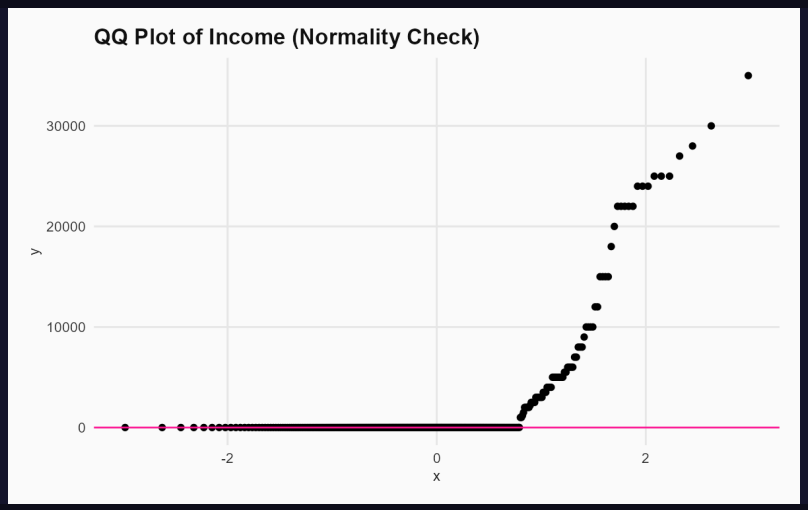
Income is heavily zero-inflated (>50% report zero). Log-transformation (log1p) was applied for modelling.



Income Density



Income vs Education



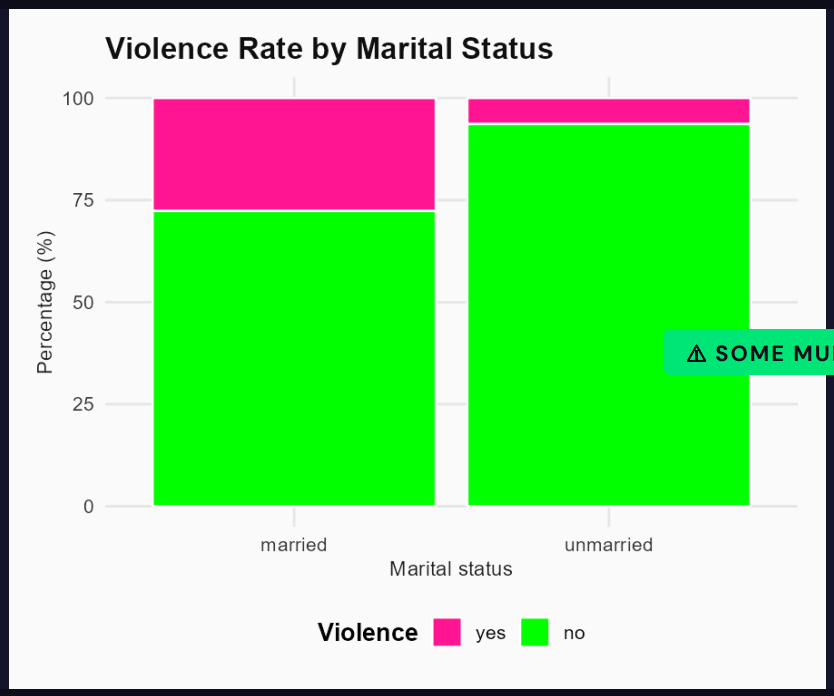
Q-Q Plot

Bimodal density with a huge spike at zero, classic zero-inflation. Violence group shows more zero-income observations.

The Q-Q plot departs strongly from normal at the tails, confirming why Wilcoxon (non-parametric) test was used.

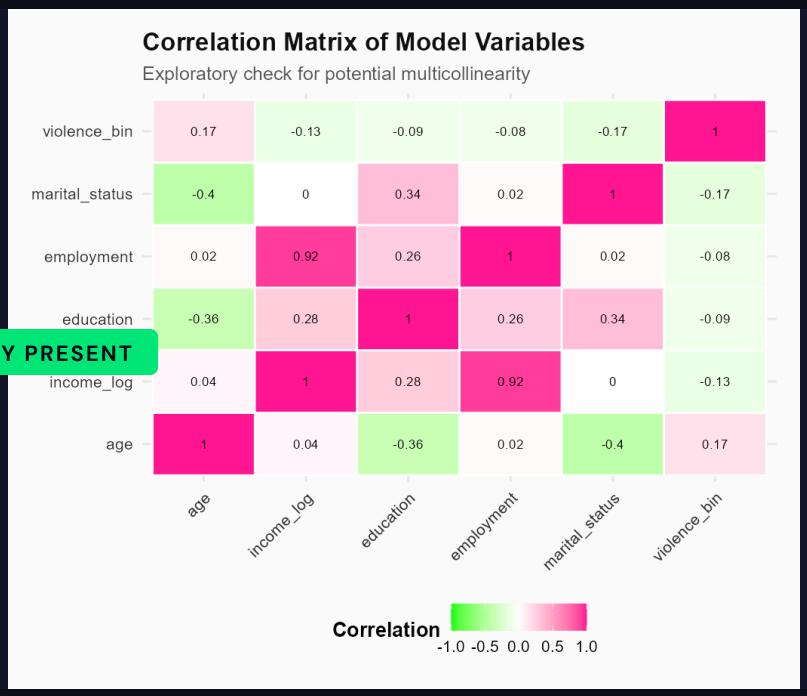
Income rises with education level across both groups, reinforcing the socioeconomic gradient in violence risk.

Marital Status & Correlations



Marital Status vs Violence

⚠ SOME MULTICOLLINEARITY PRESENT



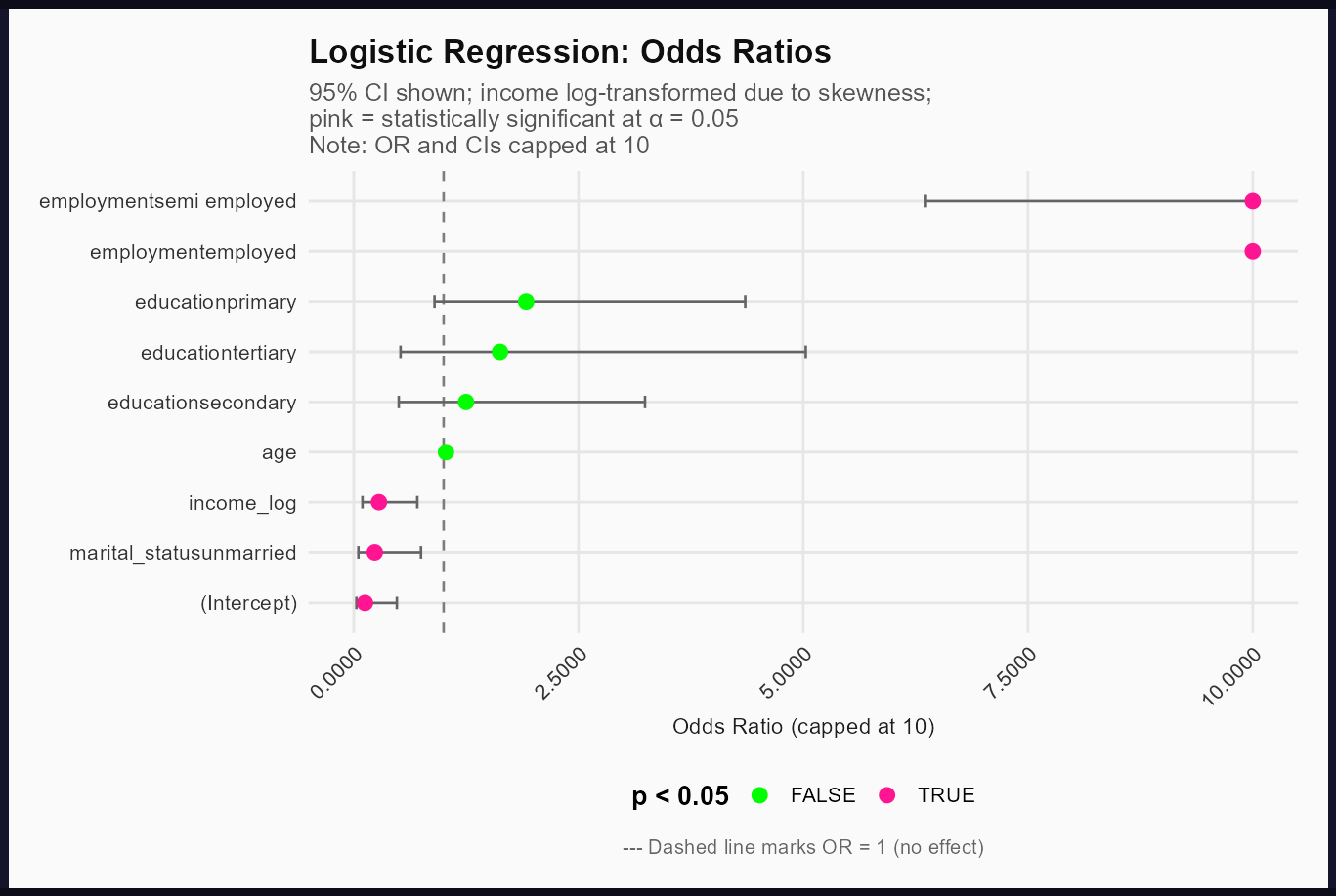
Correlation Matrix

Marital status is one of the strongest predictors. Certain statuses carry substantially elevated risk, confirmed by chi-square testing.

Most predictors show low collinearity, but income and employment are highly correlated. This overlap should be considered when interpreting their individual effects.

Odds Ratios — Forest Plot

After full adjustment, employment and marital status emerge as the strongest predictors of violence risk.



Employment

Strongest predictor: employment categories show very large ORs vs unemployed baseline

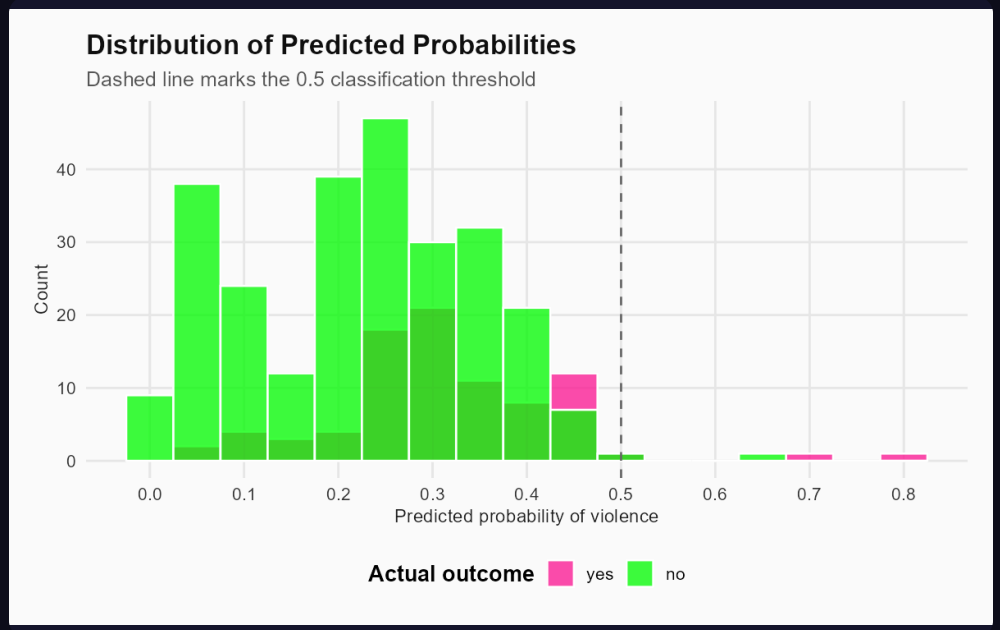
Marital Status

Significant variation across relationship types

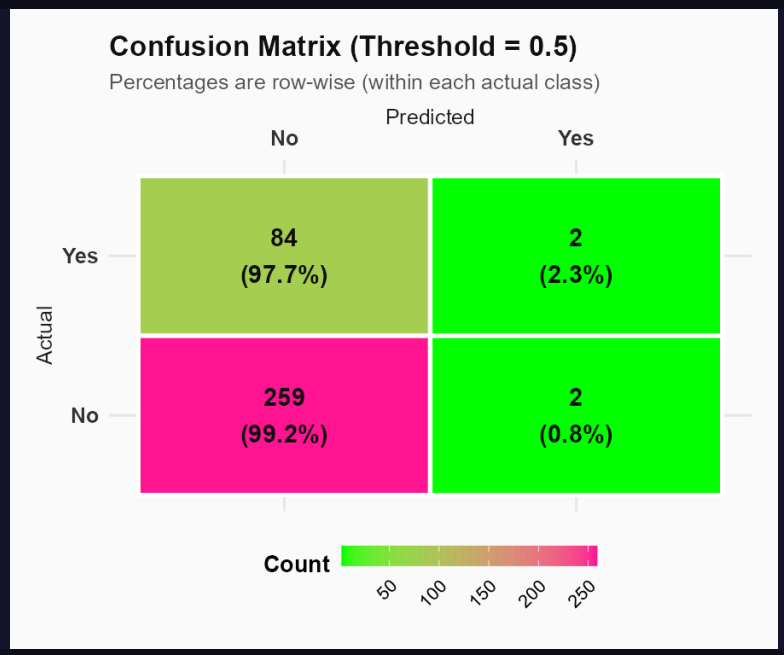
Income (log)

Higher income (financial independence) → lower violence risk

Model Performance



Predicted Probabilities



Confusion Matrix

High Specificity

Correctly identifies non-violence cases, the majority class is well handled.

Lower Sensitivity

Identifies very few true violence cases (very low sensitivity), driven by class imbalance.

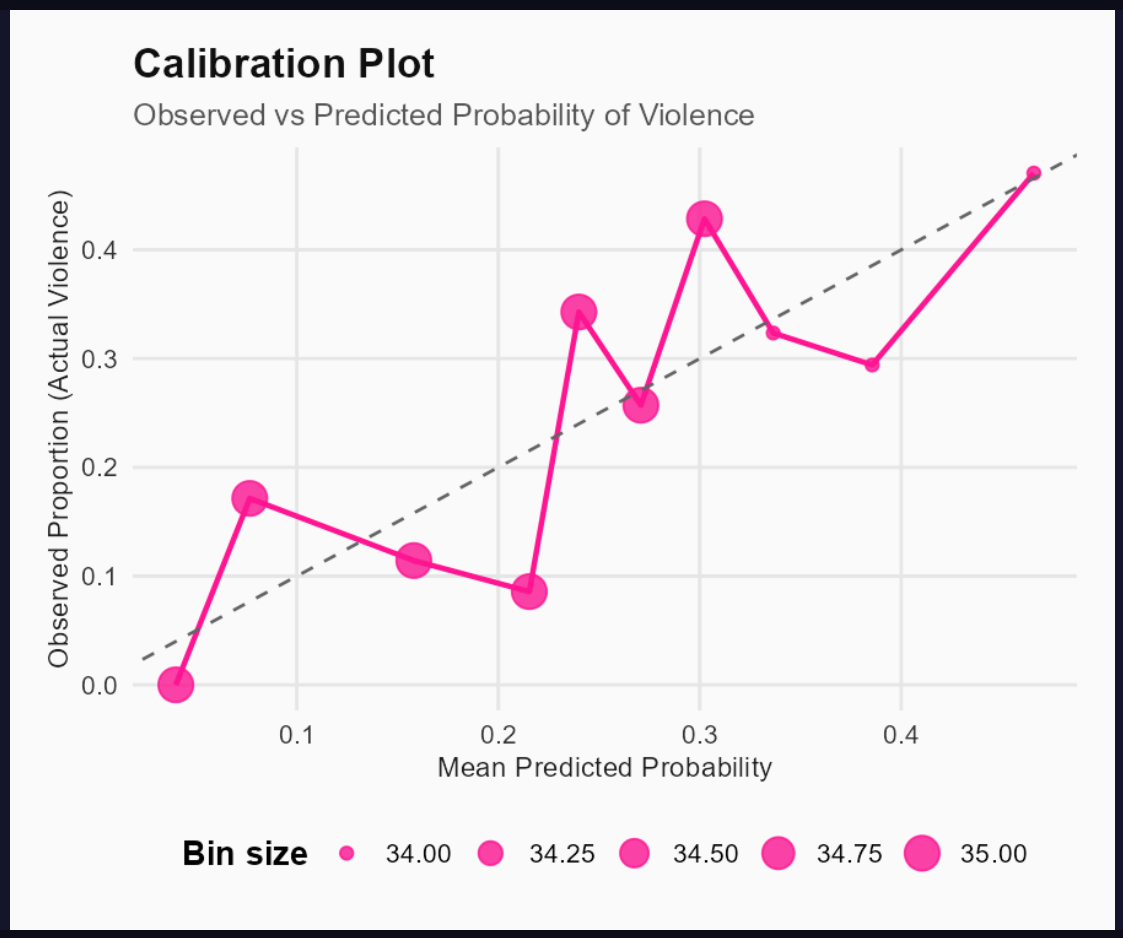
Inflated Accuracy

Overall accuracy overstated due to class distribution.

CALIBRATION

Model Calibration

Do predicted probabilities match observed rates? Points near the 45° line = well-calibrated.



45° Reference Line

Perfect calibration benchmark: predicted = observed probability.

Model Behaviour

No systematic over- or under-estimation detected.

Bin Reliability

Larger bins give more stable probability estimates.

Key Takeaways

01 Employment & marital status are the strongest and most consistent predictors of domestic violence risk in this sample.

02 Higher income (log-transformed) is significantly associated with reduced odds of violence, suggesting financial independence as a protective factor.

03 Some multicollinearity exists (notably income vs employment), requiring cautious interpretation.

04 Model is reasonably well-calibrated despite class imbalance affecting sensitivity.

05 Socioeconomic interventions (employment support, income security, education access) may represent meaningful pathways to reducing domestic violence risk.

Full code, data, and visualisations available at:

🔗 `github.com/cocobutterx13/Domestic-Violence-Analysis`

R · ggplot2 · Logistic Regression · Statistical Inference